

How Low Can Chronos-2 Go? Post-Training Quantization of a Time-Series Foundation Model

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Abstract

Time-series foundation models (TSFMs) such as Chronos-2 deliver strong zero-shot probabilistic forecasts, but their deployment cost is rarely studied and, unlike large language models, *nothing* is known about how they respond to post-training quantization (PTQ). We present the first systematic PTQ study of a TSFM: eighteen weight-quantization configurations spanning six method families (round-to-nearest, torchao, bitsandbytes INT8/NF4, HQQ, GPTQ, and activation-aware scale folding) and two- to eight-bit precision, evaluated zero-shot on the 27-dataset Chronos benchmark. Beyond point (MASE) and probabilistic (weighted quantile loss, WQL) accuracy, we introduce two calibration-aware diagnostics—mean absolute coverage error and a quantile-crossing rate—because a probabilistic forecaster can degrade in ways aggregate accuracy cannot see. Four findings emerge. (i) 8-bit quantization is free for every method ($\leq 1.3\%$ WQL change) and shrinks the model $4\times$. (ii) Hessian-aware GPTQ retains full-precision accuracy down to **3 bits** (WQL retention 1.05, 95% CI [0.99, 1.14]), where the naive baseline degrades 70%. (iii) The *method* matters far more than the *bit-width*: at 4 bits, retention spans 0.996–1.66 across methods. (iv) Interval calibration and quantile monotonicity degrade *before* accuracy does—8-bit quantization leaves WQL untouched yet quadruples the quantile-crossing rate—so accuracy-only evaluation overstates the quality of a quantized forecaster. We release the framework, which reproduces published per-task benchmark numbers to $< 10^{-6}$, and all per-series predictions.

1 Introduction

Time-series foundation models (TSFMs)—Chronos [1], Chronos-2 [2], TimesFM, Moirai—forecast unseen series zero-shot and are increasingly deployed on CPU-only servers, edge devices, and high-throughput batch pipelines where full-precision inference is costly. Quantization is the standard efficiency lever for large language models (LLMs), but its lessons do not obviously transfer to TSFMs, which are small (Chronos-2 has 120M parameters, a regime where quantization bites harder [9]), encoder-only and non-autoregressive (no KV cache, so LLM decode-centric methods do not apply), and—critically—emit *probabilistic* forecasts whose calibration and quantile monotonicity matter, not just point accuracy.

Despite an extensive LLM-quantization literature, **no published work quantizes the weights of any Chronos-family or comparable TSFM** (in that literature “quantization” usually denotes Chronos-1’s input value-*binning*, a distinct notion). We close that gap with a controlled study across method families and bit-widths, and argue that evaluating a *quantized probabilistic* forecaster demands metrics beyond those on TSFM leaderboards.

Contributions.

- The first systematic PTQ study of a TSFM: 18 configura-

tions across 6 method families and 5 bit-widths, zero-shot on 27 datasets, with bootstrap confidence intervals on every aggregate.

- A calibration-aware evaluation protocol for compressed probabilistic forecasters, adding interval-coverage error and a quantile-crossing rate, which reveal failure modes invisible to WQL/MASE.
- Findings and a deployment recommendation: GPTQ reaches full-precision accuracy *and* calibration at 3–4 bits, whereas simpler methods trade calibration first.
- An open, reproducible framework that reproduces published benchmark numbers to floating-point precision and releases all per-series predictions.

2 Setup

Model. Chronos-2 [2] (amazon/chronos-2) is a 119.5M-parameter, T5-derived *encoder-only* transformer (12 layers, $d_{\text{model}}=768$, RoPE, alternating “time” and “group” attention). It ingests patches of a per-instance-normalized series through a residual embedding (no input tokenization) and emits, in one non-autoregressive forward pass, direct multi-step forecasts at 21 quantile levels. It ships in fp32 (478 MB). All quantization is applied to the model’s `nn.Linear` weights; LayerNorm, the input embedding, and the quantile head follow standard practice and stay in fp32, and bf16 (not fp16, which overflows in T5-lineage models) is the deployment baseline.

Methods. We evaluate weight-only PTQ across six families runnable on commodity hardware: **RTN** (symmetric round-to-nearest, per-channel/ per-tensor, plus a simulated sub-8-bit sweep for the accuracy-bit curve); **torchao** INT8 weight-only and INT8 dynamic-activation (W8A8); **bitsandbytes** LLM.int8() [4] and NF4 [5]; **HQQ** [3], a calibration-free half-quadratic solver, at 8/4/3 bits; **GPTQ** [7], a Hessian-aware error-compensating quantizer we implement directly on the encoder’s linear layers with one-shot calibration; and **AWQ-style** activation-aware scale folding [8]. Methods are exercised at W8, W4 (group size 64/128), W3, and W2.

Protocol & metrics. We evaluate zero-shot on the 27-dataset Chronos benchmark [1] (190,674 forecasts) using the `fev` library—the harness the Chronos-2 authors use—so our numbers are directly comparable with the public leaderboard. Point accuracy is MASE and probabilistic accuracy is weighted quantile loss (WQL, 9 levels); per task each is normalized by a seasonal-naive baseline and aggregated by geometric mean into a *skill score*, with pairwise *win rates*. Our full-precision Chronos-2 attains 0.425 WQL skill and an 88% win rate over seasonal-naive, reproducing the paper’s ordering; as a reproduction gate, our seasonal-naive matches the published per-task MASE/WQL to $< 10^{-6}$.

For quantized models the key quantity is **retention**: the

Table 1: Representative variants, full 27-task benchmark. Retention = geometric-mean WQL(quant)/WQL(fp32), [95% bootstrap CI]. *c80* = empirical coverage of the nominal-80% interval (fp32: 0.750). *QCR* = quantile-crossing rate (fp32: 0.039). BPW = effective bits/weight incl. scales; † accuracy-exact simulation (dequantized grid in fp32)—BPW is the *achievable* footprint.

Variant	Bits	BPW	WQL ret. [CI]	MASE	c80	QCR
fp32 (ref)	32	32.0	1.000	1.000	.750	.039
RTN W8	8	8.03	0.999 [.997,1.00]	1.001	.749	.149
GPTQ W8	8	8.13†	1.000 [.990,1.01]	1.003	.749	.064
HQQ W8	8	9.00	1.002 [1.00,1.00]	1.000	.746	.086
GPTQ W4	4	4.13†	1.002 [.985,1.02]	1.011	.756	.254
HQQ W4	4	5.00	1.060 [1.02,1.12]	1.027	.630	.420
NF4 W4	4	4.13	1.065 [1.00,1.15]	1.026	.686	.396
AWQ W4	4	4.03†	1.639 [1.38,2.01]	1.409	.481	.599
GPTQ W3	3	3.25†	1.047 [.988,1.14]	1.059	.791	.366
HQQ W3	3	4.20	1.229 [1.09,1.48]	1.143	.666	.589
RTN W2	2	2.0†	4.082 [3.08,5.57]	3.108	.060	.523

geometric-mean ratio WQL(quant)/WQL(fp32) over the 27 tasks (1.00 = parity), with a 1000-resample bootstrap 95% CI. Because a probabilistic forecaster can lose quality without moving WQL, we add two diagnostics. **MACE** (mean absolute coverage error) averages $|\text{empirical} - \text{nominal}|$ over the central 80/60/40/20% intervals. **QCR** (quantile-crossing rate) is the fraction of forecast points whose predicted quantiles are non-monotone—a structural defect a direct quantile head can acquire under quantization. All per-series predictions are persisted so any further metric needs no model re-run.

3 Results

Table 1 summarizes representative configurations; Fig. 1 plots the full accuracy–bit curve and calibration. The complete 18-variant matrix is released.

F1—8-bit is free, for every method. All six W8 configurations retain WQL within [0.999, 1.013] and MASE within [1.000, 1.011] of fp32 (Table 1; Fig. 1a), at 4× compression. Even W8A8 with dynamic-activation quantization costs only 1.3%. Eight-bit quantization of Chronos-2 is, for practical purposes, lossless regardless of method.

F2—GPTQ reaches full-precision accuracy at 3 bits. GPTQ retains parity at 4 bits (retention 1.00, both group sizes; CIs include 1.0) and, remarkably, at **3 bits** (1.047, CI [0.988, 1.137]—statistically indistinguishable from fp32), where HQQ loses 23% and naive RTN 70% (Fig. 1a). Hessian-aware error compensation moves the accuracy cliff at least one bit lower than any calibration-free method. (A dev-subset signal that GPTQ *beat* fp32 did not survive the full 27-task evaluation, resolving to parity—we report the confirmed result.)

F3—method dominates bit-width. At a fixed 4 bits, WQL retention spans 0.996 (GPTQ) to 1.66 (RTN)—a 66-point gap—whereas moving a *good* method from 8 to 4 bits costs $\leq 6\%$ (Fig. 1a). For a TSFM, *how* you quantize matters far more than *how much*, inverting the intuition that bit-width is the primary knob.

Table 2: Inference efficiency (storage-real variants, NVIDIA RTX 2000 Ada laptop, context 2048, horizon 64). Peak mem and latency at batch 1; throughput at batch 256.

Model	Size MB	Peak GPU MB (b1)	GPU lat. ms (b1)	Thr. ser./s
fp32	478	493	30	110
INT8 (HQQ)	120	176	42	122
NF4 (W4)	60	87	45	120

F4—calibration and monotonicity degrade before accuracy, and only GPTQ preserves them. This is the study’s central cautionary finding. At 8 bits, accuracy is perfectly retained yet the quantile-crossing rate already quadruples (0.039 $\rightarrow$ 0.149 for RTN)—a defect in the quantile head that WQL and MASE cannot see. At 4 bits, GPTQ keeps interval coverage intact (c80 0.756 vs fp32 0.750; Fig. 1b) while HQQ and NF4 sag to 0.630 and 0.686—a user trusting the “80%” band of HQQ-W4 actually holds a 63% band. GPTQ is the *only* 4-bit method preserving both accuracy and calibration; no method fully preserves quantile monotonicity. Evaluating a quantized probabilistic forecaster on accuracy alone therefore overstates its deployable quality [6].

F5—damage is tail-concentrated. Mean retention hides worst cases. HQQ-W3 averages 1.23 but its worst task is $8.2\times$ and 44% of tasks exceed 1.10; GPTQ-W4 has a median per-task ratio of 0.998 and only 4% of tasks above 1.10. Per-series distributions—released for every run—are the honest picture, echoing the “hard inputs degrade first” pattern from encoder-model quantization.

F6—activation-aware folding is a rigorous negative result. For Chronos-2 the AWQ scale fold is *exact* (pre-norm RMSNorm weights absorb the scale; the ReLU MLP is positively homogeneous; $v \rightarrow o$ folds cleanly), which we verify numerically. Yet at 4 bits AWQ-fold+RTN (1.64) barely differs from plain RTN (1.66) and is far behind GPTQ (Fig. 1a). On this model, error *compensation* (GPTQ) is the effective mechanism; activation *scaling* alone is not.

Efficiency. Storage-real 4-bit models occupy ≈ 60 MB (8× smaller) and cut peak batch-1 GPU memory from 493 MB (fp32) to 87 MB for NF4 (5.7×; Table 2). Throughput is flat (~ 120 series/s at batch 256, vs 110 for fp32): our reference dequantize-then-GEMM paths buy footprint, not speed—the expected outcome for unfused weight-only kernels, and why optimized-kernel or activation-quantized paths are needed for latency. Memory-bound and edge deployments benefit immediately.

4 Discussion, Limitations, Conclusion

Recommendation. To deploy Chronos-2: use 8-bit for a free 4× reduction with any toolkit; for 4 bits and below, use GPTQ (or another Hessian-aware method) rather than a calibration-free quantizer—not merely for accuracy but because it is the only method that preserves interval calibration, the property forecasting users depend on. Report calibration and quantile-crossing alongside WQL whenever a probabilistic model is

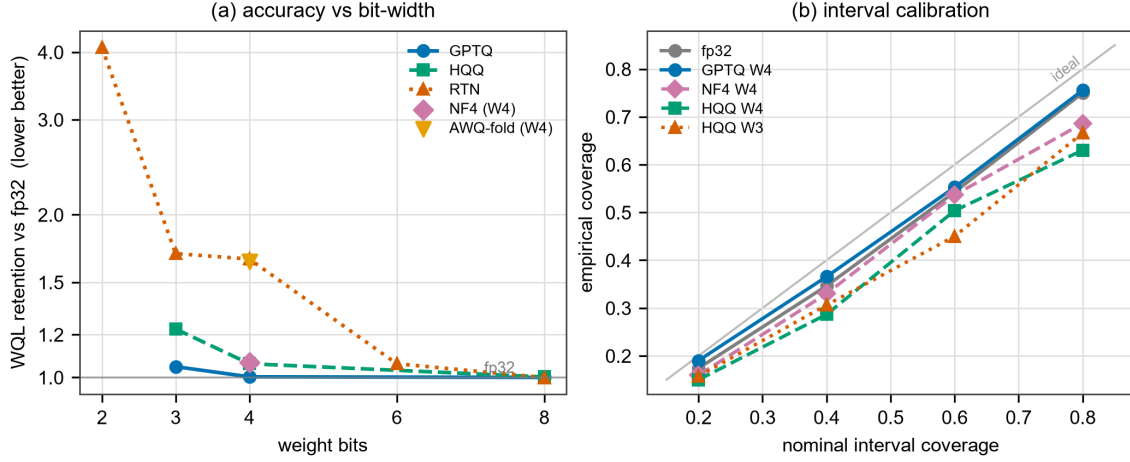


Figure 1: (a) WQL retention vs weight bits by method (log scale; lower is better, 1.0 = fp32). GPTQ tracks fp32 to 3 bits; the method spread at 4 bits exceeds the 8→4-bit gap of any single method; AWQ-fold sits on the RTN line. (b) Empirical vs nominal interval coverage: GPTQ-W4 tracks fp32 while HQQ/NF4 systematically under-cover.

compressed.

Limitations. We study one 120M model on univariate zero-shot tasks; covariate/multivariate tasks, other TSFM sizes, and quantization-aware training are future work. Our GPTQ/AWQ arms are accuracy-exact simulations (dequantized grids in fp32 containers); realized latency for them requires packed kernels we do not implement (achievable footprint is reported). Latency is hardware-specific; retention ratios are the transferable claim. We assess but exclude KV-cache/vector methods such as TurboQuant [10], which do not apply to a cache-free encoder.

Conclusion. Chronos-2 quantizes remarkably well: 8-bit is free, and Hessian-aware 4- and 3-bit quantization retains full-precision forecasting accuracy. But probabilistic structure—interval calibration and quantile monotonicity—degrades before accuracy, and only error-compensating methods preserve it. Compressing a probabilistic forecaster is therefore not merely an accuracy-retention problem, and should not be evaluated as one. Code, configs, per-task summaries, and all per-series predictions are released; every number is regenerated by one command.

References

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